CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

L&T EduTech
Advanced JavaScript Frameworks (Angular)
CIA 3 – Projects Activities

Submitted by:

BAREEQ QAYOOM - 2460343

SHREYA SUNIL MORAJKAR - 2460450

JOHN JOBY C - 2460380

NAVANEET NAIR - 2460409

Employee Management Dashboard using Angular and TypeScript

Abstract

This project presents the development of an Employee Management Dashboard built using the Angular framework and TypeScript. The application demonstrates a modular and component-based architecture for managing employee data efficiently. It provides an interactive web interface that allows users to view, add, and manage employee records. The system integrates Angular Material components to provide a responsive and professional user interface.

The project highlights the use of Angular's key features such as components, services, routing, reactive forms, dependency injection, pipes, and directives. By simulating CRUD (Create, Read, Update, Delete) operations, the dashboard provides a practical example of modern front-end development practices. The application is designed to be scalable, maintainable, and adaptable for future integration with backend APIs.

---

1. Introduction

Modern organizations require efficient digital systems to manage employee information and organizational data. Traditional methods such as spreadsheets or manual record keeping often lead to inconsistencies, lack of validation, and inefficient data handling. Web-based dashboards provide a more efficient solution by enabling centralized data management and real-time interaction.

Angular is a powerful TypeScript-based front-end framework developed by Google. It supports component-based architecture and reactive programming, making it suitable for building scalable enterprise applications.

This project implements an Employee Management Dashboard using Angular. The application demonstrates how Angular components, services, routing, and reactive forms can be combined to build a responsive user interface for managing employee data. Angular Material is integrated to provide modern UI components and ensure a professional look and feel.

---

2. Problem Statement

Organizations need efficient systems to manage employee records, including personal details, roles, departments, and salary information. Traditional spreadsheet-based systems lack real-time updates, validation mechanisms, and interactive user interfaces.

Key challenges include:

1. Lack of centralized employee data management

2. Absence of real-time updates and dynamic data interaction
3. Poor validation and risk of incorrect data entry
4. Limited scalability for expanding organizational needs
5. Unstructured and non-responsive user interfaces

This project addresses these challenges by developing a lightweight yet scalable Angular-based dashboard that enables efficient employee data management.

---

3. Infrastructure Requirements

Software Requirements

- Node.js (v18 or higher)
- Angular CLI (v17 or higher)
- npm (Node Package Manager)
- Visual Studio Code IDE
- Git for version control

Framework and Libraries

- Angular Framework
- TypeScript
- Angular Material
- RxJS (Reactive Extensions)
- Angular Router
- Angular Reactive Forms

Hardware Requirements

- Processor: Dual-core 2 GHz or higher
- RAM: Minimum 8 GB
- Storage: 2 GB available space
- Internet connection for package installation

---

4. System Architecture

The application follows Angular's component-based architecture, ensuring modularity and maintainability.

Architecture Structure

```
Angular Application
│
├── Components
│   ├── EmployeeListComponent
│   ├── EmployeeDetailComponent
│   ├── EmployeeFormComponent
│   └── NavbarComponent
│
├── Services
│   └── EmployeeService
│
├── Models
│   └── Employee Interface
│
├── Pipes
│   └── DepartmentFilterPipe
│
├── Directives
│   └── HighlightSalaryDirective
│
└── Routing
    └── AppRoutes
```

Data Flow

1. User interacts with Angular UI components.
2. Components request data through the EmployeeService.
3. Services manage data operations and business logic.
4. Angular change detection updates the UI automatically.

---

5. Implementation Details

Employee Data Model

```
export interface Employee {
  id: number;
  name: string;
  role: string;
  department: string;
  salary: number;
  email?: string;
  joinDate?: Date;
}
```

Core Components

Employee List Component

- Displays all employees in a table format.
- Uses Angular Material MatTable component.
- Supports department filtering using custom pipe.
- Uses structural directive *ngFor.

Employee Detail Component

- Displays detailed employee information.
- Uses route parameters to retrieve employee data.
- Implements Angular lifecycle hook ngOnInit.

Employee Form Component

- Built using Angular Reactive Forms.
- Supports employee creation and editing.
- Includes validation for required fields and salary limits.

Navbar Component

- Provides navigation across the application.
- Implemented using Angular Material Toolbar.

---

6. Key Features Implemented

Angular Material Integration

- MatTable for displaying employee data
- MatToolbar for navigation
- MatButton for user interactions
- MatFormField and MatInput for forms

Routing

Angular Router enables navigation between pages.

Example Routes:

```
const routes = [
  { path: '', component: EmployeeListComponent },
  { path: 'employee/:id', component: EmployeeDetailComponent },
  { path: 'add-employee', component: EmployeeFormComponent }
];
```

Custom Pipe

DepartmentFilterPipe filters employees by department.

Custom Directive

HighlightSalaryDirective highlights employees with high salaries.

Reactive Forms

Employee creation form uses FormGroup and FormControl.

Validation includes:

- Required fields
- Minimum salary value
- Email format validation

---

7. Testing Strategy

Testing ensures the reliability and functionality of the application.

Unit Testing

Angular TestBed is used for testing components and services.

Form Validation Testing

Ensures validation rules work correctly.

Routing Testing

Ensures correct navigation between pages.

Service Testing

EmployeeService methods are tested for correct CRUD operations.

---

8. Deployment Considerations

The application can be deployed using Angular production build.

Production build command:

ng build --configuration production

Deployment platforms include:

- Firebase Hosting
- Netlify
- GitHub Pages
- AWS S3

Build optimization ensures:

- Minified CSS and JavaScript
- Faster page load times
- Improved application performance

---

9. Application Screenshots

Example Screenshots:

- Employee Dashboard
- Add Employee Form
- Employee Detail Page

(Screenshots can be added here in the report.)

---

10. Future Enhancements

The system can be improved with additional features such as:

- Authentication and authorization
- Integration with backend REST APIs
- Employee search and advanced filtering
- Pagination and sorting
- Export employee data to CSV or PDF
- Analytics dashboard for HR insights

Results and Output

Development Environment Setup

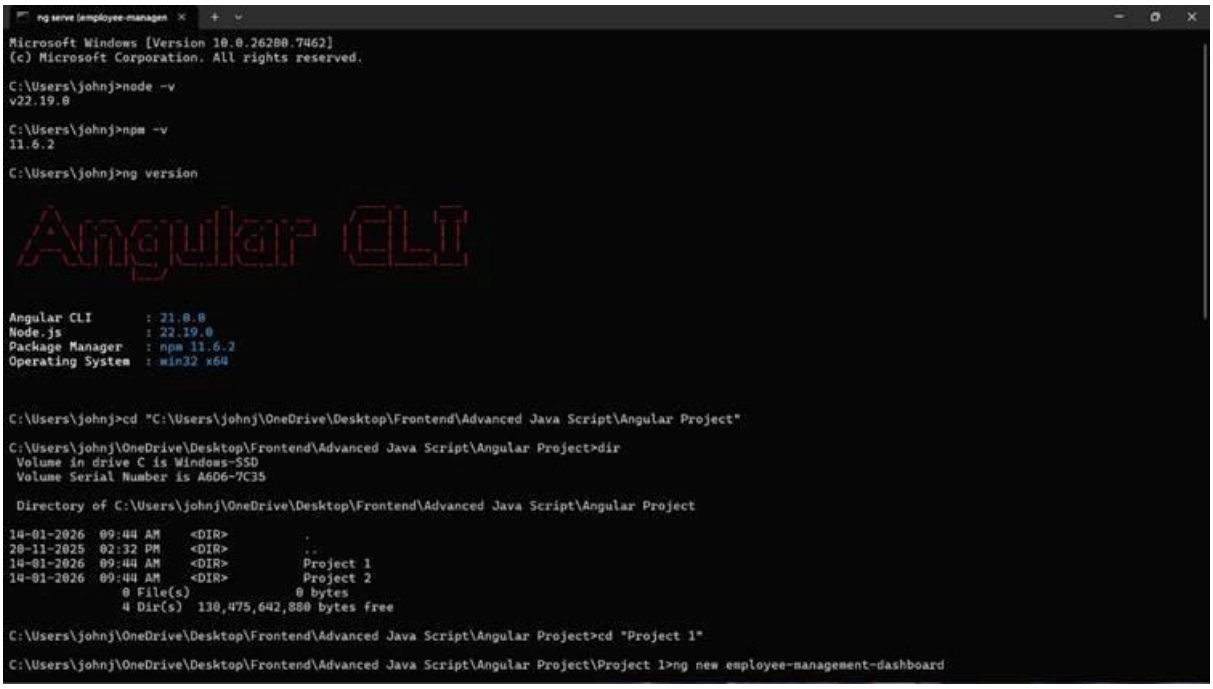


Figure 1: Environment verification showing Node.js, npm, and Angular versions

Project Initialization

```
C:\Users\johnj\OneDrive\Desktop\Frontend\Advanced Java Script\Angular Project\Project 1>ng new employee-management-dashboard
/ Which stylesheet system would you like to use? CSS              [
https://developer.mozilla.org/docs/Web/CSS                        ]
/ Do you want to enable Server-Side Rendering (SSR) and Static Site
Generation (SSG/Prerendering)? No
/ Which AI tools do you want to configure with Angular best practices?
https://angular.dev/ai/develop-with-ai None
CREATE employee-management-dashboard/angular.json (2045 bytes)
CREATE employee-management-dashboard/package.json (976 bytes)
CREATE employee-management-dashboard/README.md (1549 bytes)
CREATE employee-management-dashboard/tsconfig.json (990 bytes)
CREATE employee-management-dashboard/.editorconfig (331 bytes)
CREATE employee-management-dashboard/.gitignore (647 bytes)
CREATE employee-management-dashboard/tsconfig.app.json (444 bytes)
CREATE employee-management-dashboard/tsconfig.spec.json (456 bytes)
CREATE employee-management-dashboard/.vscode/extensions.json (134 bytes)
CREATE employee-management-dashboard/.vscode/launch.json (490 bytes)
CREATE employee-management-dashboard/.vscode/tasks.json (980 bytes)
CREATE employee-management-dashboard/src/main.ts (228 bytes)
CREATE employee-management-dashboard/src/index.html (326 bytes)
CREATE employee-management-dashboard/src/styles.css (81 bytes)
CREATE employee-management-dashboard/src/app/app.spec.ts (719 bytes)
CREATE employee-management-dashboard/src/app/app.ts (323 bytes)
CREATE employee-management-dashboard/src/app/app.css (0 bytes)
CREATE employee-management-dashboard/src/app/app.html (20464 bytes)
CREATE employee-management-dashboard/src/app/app.config.ts (324 bytes)
CREATE employee-management-dashboard/src/app/app.routes.ts (80 bytes)
CREATE employee-management-dashboard/public/favicon.ico (15086 bytes)
/ Packages installed successfully.
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
    Successfully initialized git.

C:\Users\johnj\OneDrive\Desktop\Frontend\Advanced Java Script\Angular Project\Project 1>cd employee-management-dashboard

C:\Users\johnj\OneDrive\Desktop\Frontend\Advanced Java Script\Angular Project\Project 1\employee-management-dashboard>ng serve
Initial chunk files | Names         | Raw size
main.js             | main          | 48.45 kB |
styles.css          | styles        | 95 bytes |

                    | Initial total | 48.55 kB

Application bundle generation complete. [4.508 seconds] - 2026-01-14T04:23:46.458Z
```

Figure 2: Angular CLI project creation process

Application Structure

```
CREATE employee-management-dashboard/.gitignore (647 bytes)
CREATE employee-management-dashboard/tsconfig.app.json (444 bytes)
CREATE employee-management-dashboard/tsconfig.spec.json (456 bytes)
CREATE employee-management-dashboard/.vscode/extensions.json (134 bytes)
CREATE employee-management-dashboard/.vscode/launch.json (490 bytes)
CREATE employee-management-dashboard/.vscode/tasks.json (980 bytes)
CREATE employee-management-dashboard/src/main.ts (228 bytes)
CREATE employee-management-dashboard/src/index.html (326 bytes)
CREATE employee-management-dashboard/src/styles.css (81 bytes)
CREATE employee-management-dashboard/src/app/app.spec.ts (719 bytes)
CREATE employee-management-dashboard/src/app/app.ts (323 bytes)
CREATE employee-management-dashboard/src/app/app.css (0 bytes)
CREATE employee-management-dashboard/src/app/app.html (20464 bytes)
CREATE employee-management-dashboard/src/app/app.config.ts (324 bytes)
CREATE employee-management-dashboard/src/app/app.routes.ts (80 bytes)
CREATE employee-management-dashboard/public/favicon.ico (15086 bytes)
/ Packages installed successfully.
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
    Successfully initialized git.

C:\Users\johnj\OneDrive\Desktop\Frontend\Advanced Java Script\Angular Project\Project 1>cd employee-management-dashboard

C:\Users\johnj\OneDrive\Desktop\Frontend\Advanced Java Script\Angular Project\Project 1\employee-management-dashboard>ng serve
Initial chunk files | Names         | Raw size
main.js             | main          | 48.45 kB |
styles.css          | styles        | 95 bytes |

                    | Initial total | 48.55 kB

Application bundle generation complete. [4.508 seconds] - 2026-01-14T04:23:46.458Z

Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
  → Local:   http://localhost:4200/
  → press h + enter to show help
```

Figure 3: Project structure generation and development server startup

Functional Output

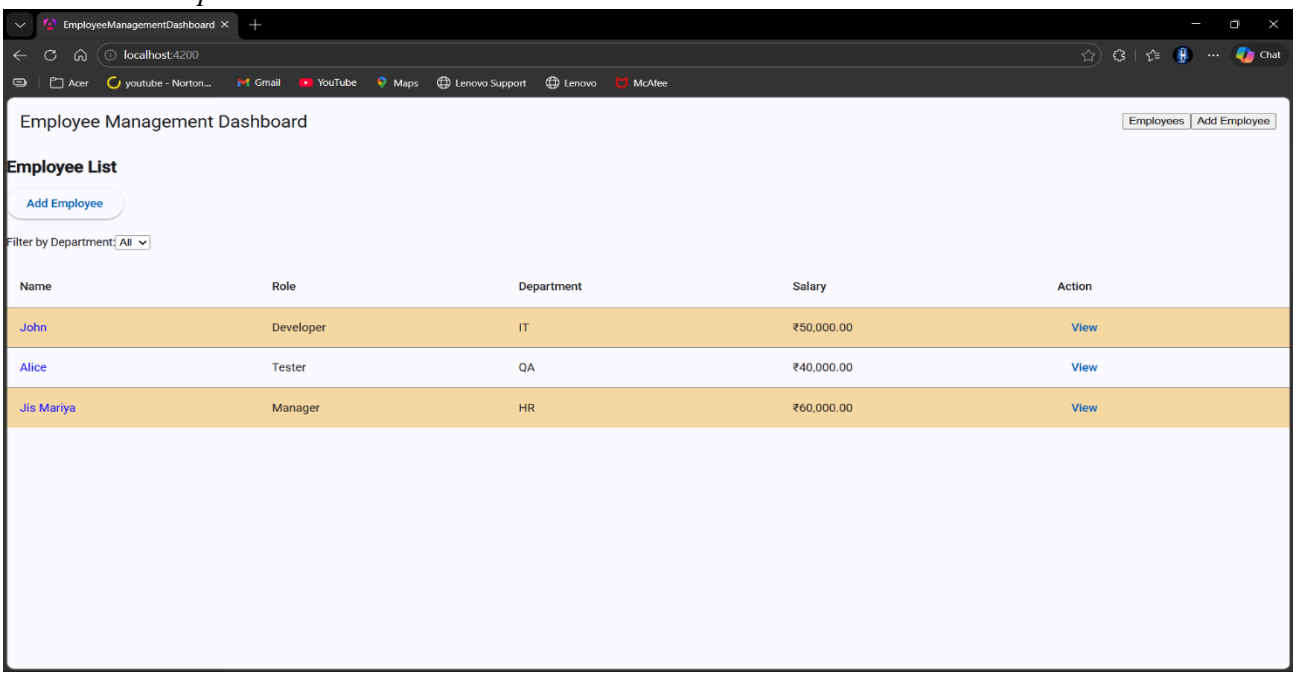


Figure 4: Employee list interface with sample data

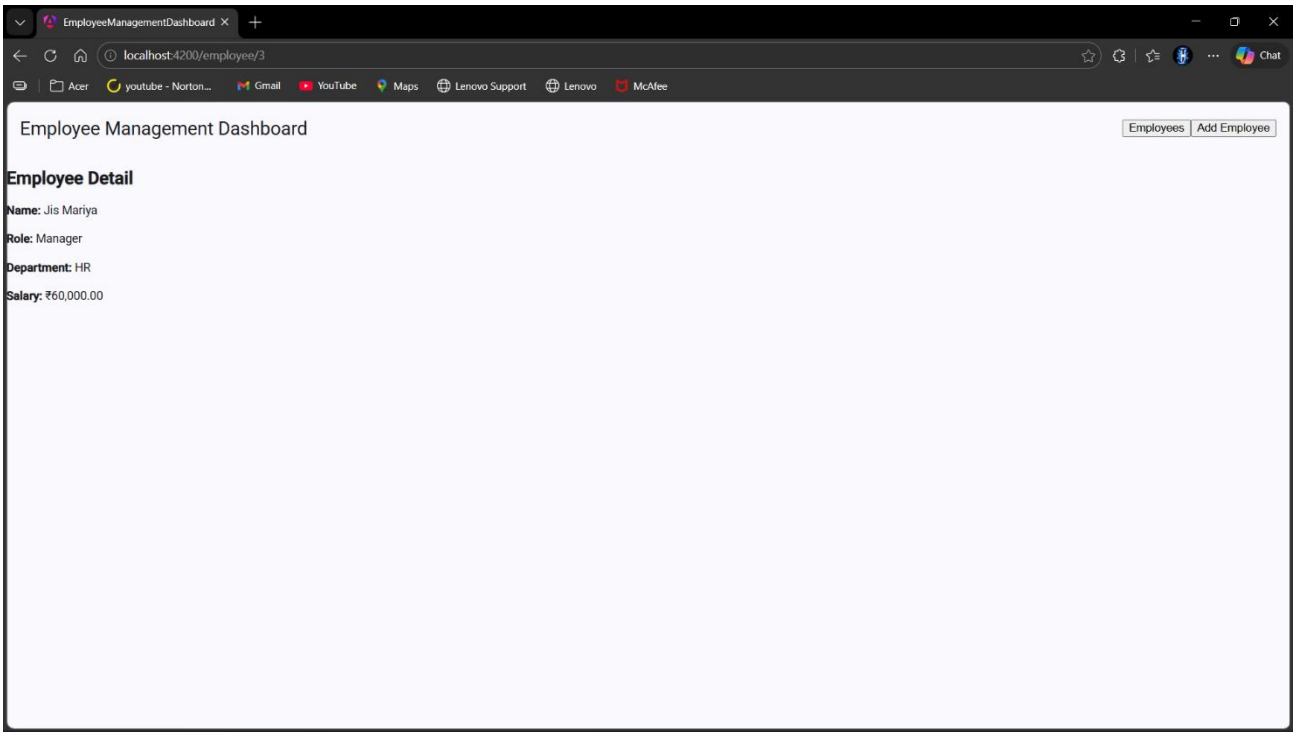
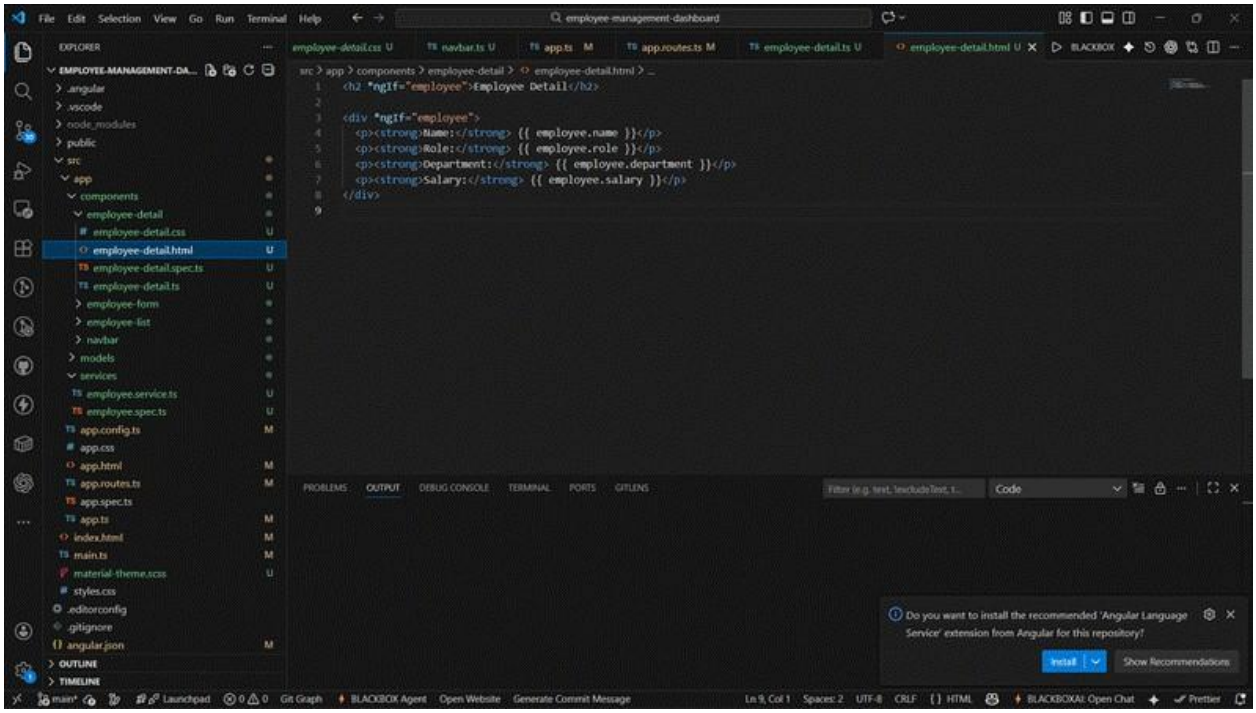
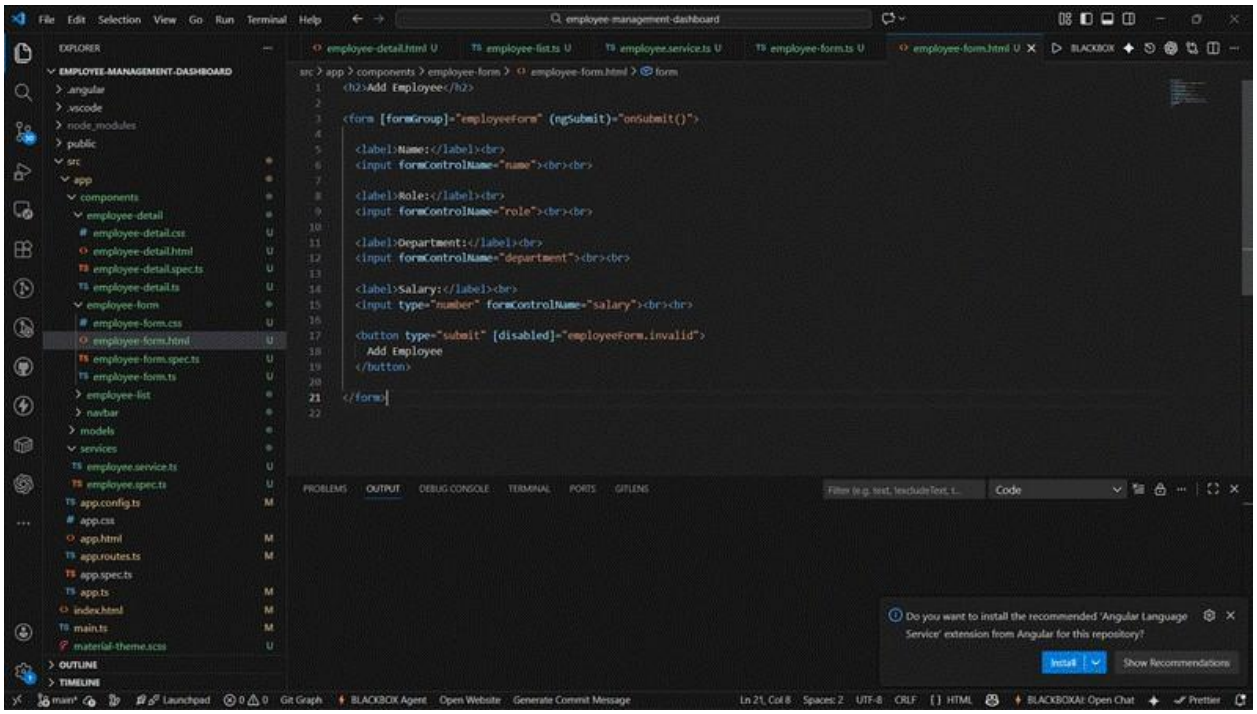


Figure 5: Employee detail view for Jis Mariya (HR)

Screenshots of Code:





File Edit Selection View Go Run Terminal Help

EXPLORER

EMPLOYEE-MANAGEMENT-DASHBOARD

- node_modules
- public
- src
  - app
    - components
      - employee-detail
        - employee-detail.css
        - employee-detail.html
        - employee-detail.spec.ts
        - employee-detail.ts
      - employee-form
        - employee-form.css
        - employee-form.html
        - employee-form.spec.ts
        - employee-form.ts
      - employee-list
        - employee-list.css
        - employee-list.html
        - employee-list.spec.ts
        - employee-list.ts
      - navbar
    - models
    - services
      - employee.service.ts
      - employee.spec.ts
    - app.config.ts
    - app.css
    - app.html
    - app.routes.ts
    - app.spec.ts
    - app.ts
    - index.html

OUTLINE

TIMELINE

employee-list.ts U | employee.service.ts U | employee-form.ts U | employee-form.html U | employee-list.html U

src > app > components > employee-list > employee-list.ts > ...

```
import { Component } from '@angular/core';
import { CommonModule } from '@angular/common';
import { Router } from '@angular/router';
import { EmployeeService } from '../../services/employee.service';

/* Angular Material imports */
import { MatTableModule } from '@angular/material/table';
import { MatButtonModule } from '@angular/material/button';

@Component({
  selector: 'app-employee-list',
  standalone: true,
  imports: [CommonModule, MatTableModule, MatButtonModule],
  templateUrl: './employee-list.html',
  styleUrls: ['./employee-list.css']
})

export class EmployeeListComponent {

  employees: any[] = [];
  displayedColumns: string[] = ['name', 'role', 'department', 'salary', 'actions'];

  constructor(
    private router: Router,
    private employeeService: EmployeeService
  ) {
    this.employees = this.employeeService.getEmployees();
  }

  viewEmployee(id: number) {
    this.router.navigate(['/employee', id]);
  }
}
```

Do you want to install the recommended 'Angular Language Service' extension from Angular for this repository?

Install | Show Recommendations

Launchpad | Git Graph | BLACKBOX Agent | Open Website | Generate Commit Message | Ln 34, Col 1 | Spaces: 2 | UTF-8 | CRLF | TypeScript | BLACKBOXAI: Open Chat | Prettier

File Edit Selection View Go Run Terminal Help

EXPLORER

EMPLOYEE-MANAGEMENT-DASHBOARD

- node_modules
- public
- src
  - app
    - components
      - employee-detail
        - employee-detail.css
        - employee-detail.html
        - employee-detail.spec.ts
        - employee-detail.ts
      - employee-form
        - employee-form.css
        - employee-form.html
        - employee-form.spec.ts
        - employee-form.ts
      - employee-list
        - employee-list.css
        - employee-list.html
        - employee-list.spec.ts
        - employee-list.ts
      - navbar
        - navbar.css
        - navbar.html
        - navbar.spec.ts
        - navbar.ts
    - models
    - services
      - employee.service.ts
      - employee.spec.ts
    - app.config.ts
    - app.css
    - app.html

OUTLINE

TIMELINE

navbar.ts U | app.ts M | app.routes.ts M | employee-detail.ts U | employee-detail.html U | employee-list.ts U

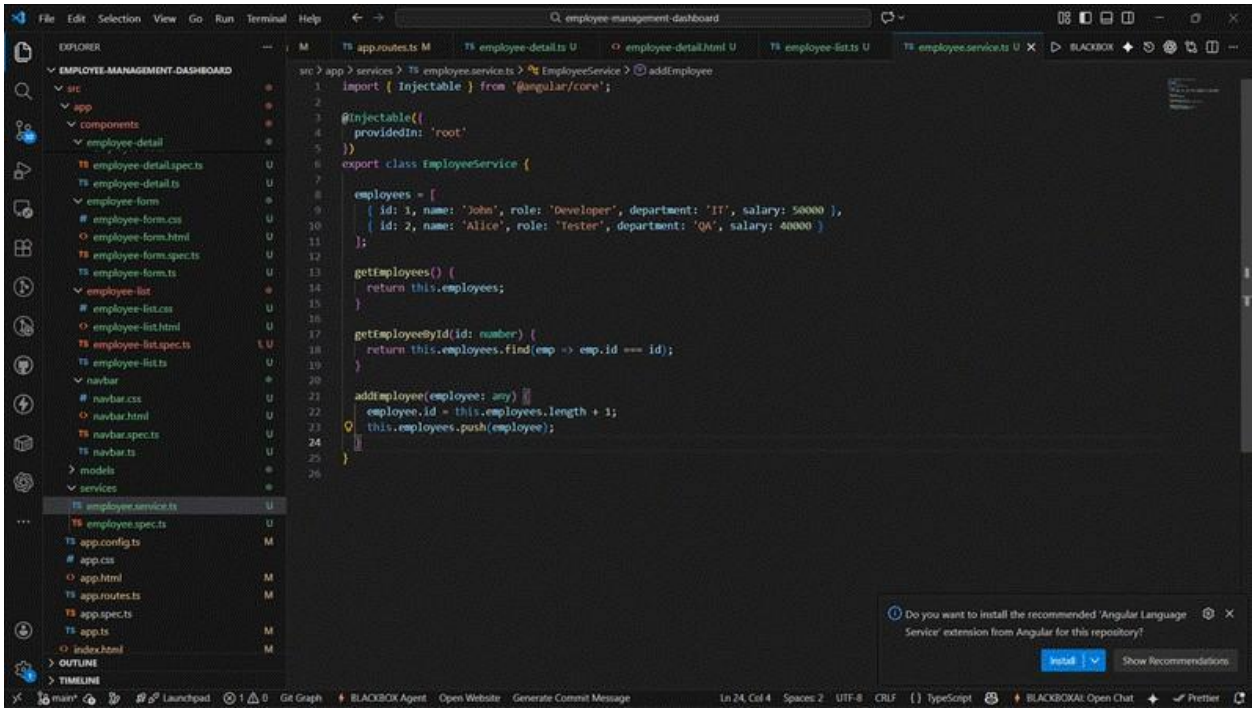
src > app > components > navbar > navbar.ts > ...

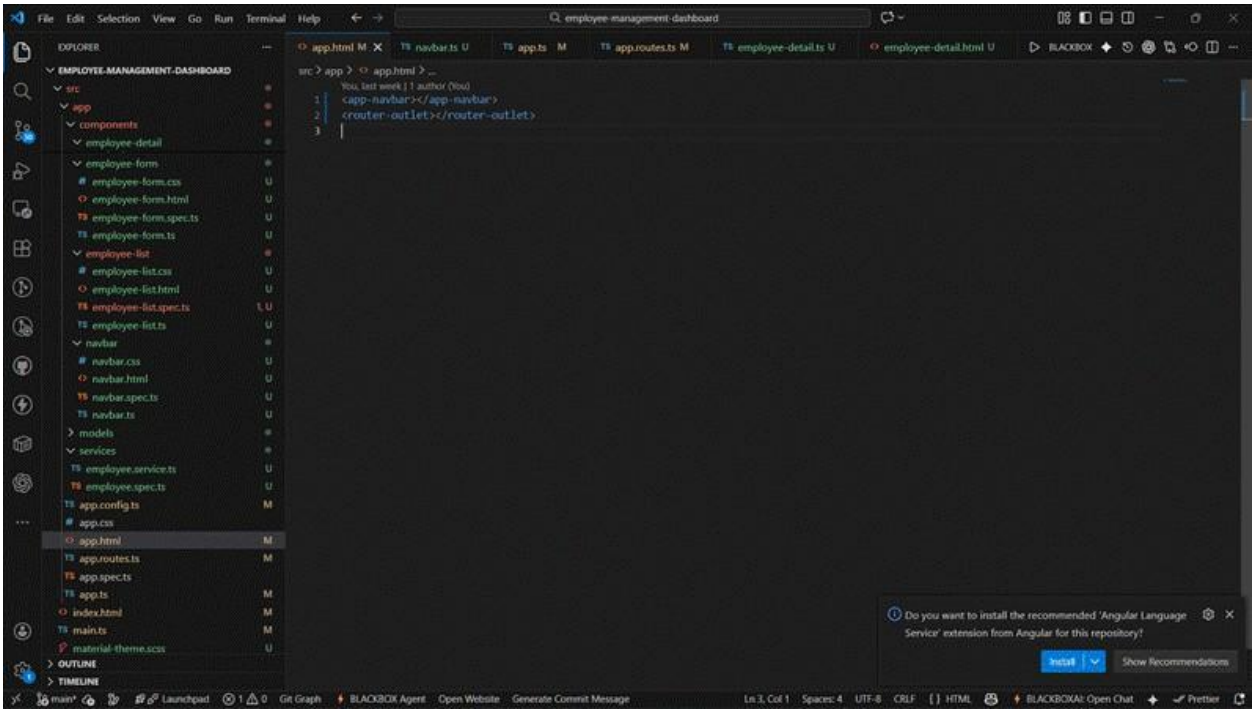
```
import { Component } from '@angular/core';
import { CommonModule } from '@angular/common';
import { MatToolbarModule } from '@angular/material/toolbar';

@Component({
  selector: 'app-navbar',
  standalone: true,
  imports: [CommonModule, MatToolbarModule],
  templateUrl: './navbar.html',
  styleUrls: ['./navbar.css']
})
export class NavbarComponent {}
```

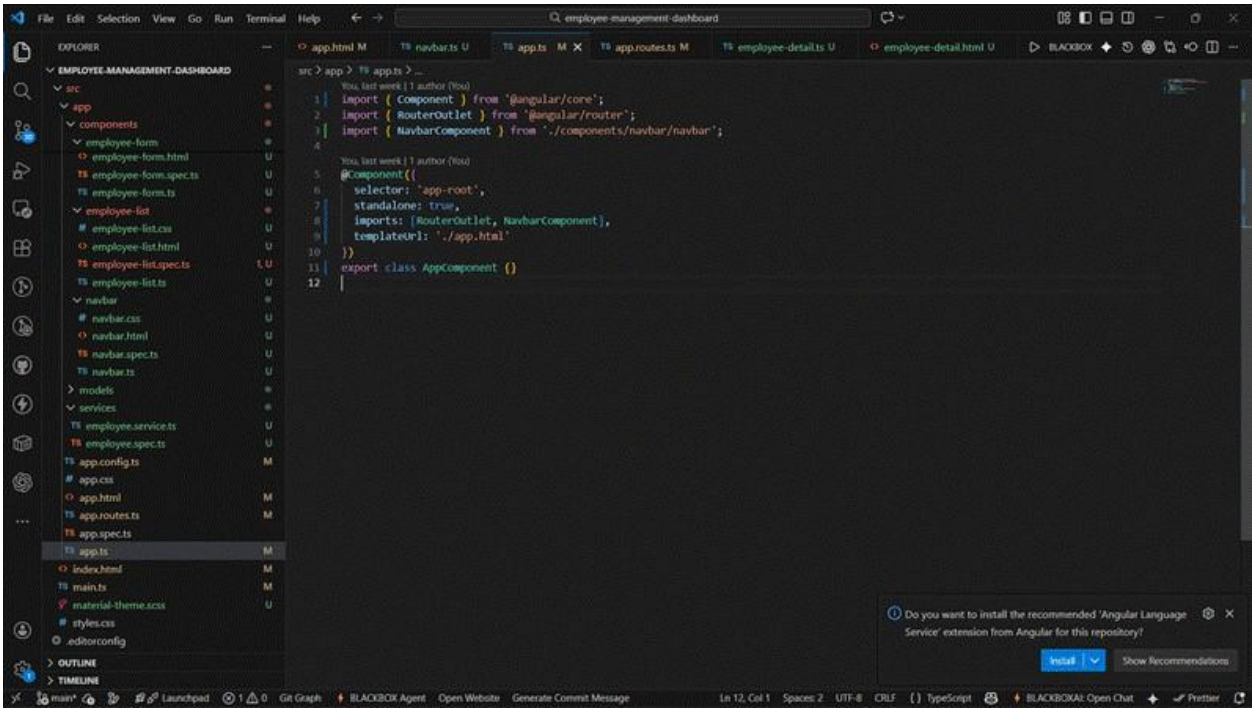
Do you want to install the recommended 'Angular Language Service' extension from Angular for this repository?

Install | Show Recommendations

Launchpad | Git Graph | BLACKBOX Agent | Open Website | Generate Commit Message | Ln 13, Col 1 | Spaces: 2 | UTF-8 | CRLF | TypeScript | BLACKBOXAI: Open Chat | Prettier


File Edit Selection View Go Run Terminal Help
employee-management-dashboard
EXPLORER
EMPLOYEE-MANAGEMENT-DASHBOARD
src
app
components
employee-detail
employee-detail.spec.ts
employee-detail.ts
employee-form
employee-form.css
employee-form.html
employee-form.spec.ts
employee-form.ts
employee-list
employee-list.css
employee-list.html
employee-list.spec.ts
employee-list.ts
navbar
navbar.css
navbar.html
navbar.spec.ts
navbar.ts
models
services
employee.service.ts
employee.spec.ts
app.config.ts
app.css
app.html
app.routes.ts
app.spec.ts
app.ts
index.html
OUTLINE
TIMELINE
app.routes.ts M
employee-detail.ts U
employee-detail.html U
employee-list.ts U
employee.service.ts U
src > app > services > employee.service.ts > EmployeeService > addEmployee
import { Injectable } from '@angular/core';
@Injectable({
providedIn: 'root'
})
export class EmployeeService {
employees = [
{ id: 1, name: 'John', role: 'Developer', department: 'IT', salary: 50000 },
{ id: 2, name: 'Alice', role: 'Tester', department: 'QA', salary: 40000 }
];
getEmployees() {
return this.employees;
}
getEmployeeById(id: number) {
return this.employees.find(emp => emp.id === id);
}
addEmployee(employee: any) {
employee.id = this.employees.length + 1;
this.employees.push(employee);
}
}
Do you want to install the recommended 'Angular Language Service' extension from Angular for this repository?
Install
Show Recommendations
Ln 24, Col 4 Spaces: 2 UTF-8 CRLF {} TypeScript
Prettier


File Edit Selection View Go Run Terminal Help
employee-management-dashboard
EXPLORER
EMPLOYEE-MANAGEMENT-DASHBOARD
src
app
components
employee-detail
employee-form
employee-form.css
employee-form.html
employee-form.spec.ts
employee-form.ts
employee-list
employee-list.css
employee-list.html
employee-list.spec.ts
employee-list.ts
navbar
navbar.css
navbar.html
navbar.spec.ts
navbar.ts
models
services
employee.service.ts
employee.spec.ts
app.config.ts
app.css
app.html
app.routes.ts
app.spec.ts
app.ts
index.html
main.ts
material-theme.scss
OUTLINE
TIMELINE
app.html M
navbar.ts U
app.ts M
app.routes.ts M
employee-detail.ts U
employee-detail.html U
src > app > app.html > ...
You, last week | 1 author (You)
<app-navbar></app-navbar>
<router-outlet></router-outlet>
Do you want to install the recommended 'Angular Language Service' extension from Angular for this repository?
Install
Show Recommendations
Ln 3, Col 1 Spaces: 4 UTF-8 CRLF {} HTML
Prettier



Conclusion

The Employee Management Dashboard successfully demonstrates the practical implementation of Angular for building modern web applications. By combining component-based architecture, reactive forms, routing, and Angular Material UI components, the project provides a scalable and maintainable solution for managing employee information.

The project highlights key Angular development concepts such as dependency injection, custom pipes, directives, and reactive programming. It serves as a strong foundation for future enhancements including backend integration and enterprise-level features.

Overall, the application showcases how Angular can be used effectively to develop professional, responsive, and scalable front-end applications.

GitHub

https://github.com/JohnJoby2006/L-T-Project-Employee-Mana gement-Project.git

*You need to run the code in command Prompt by typing “ng serve -o”